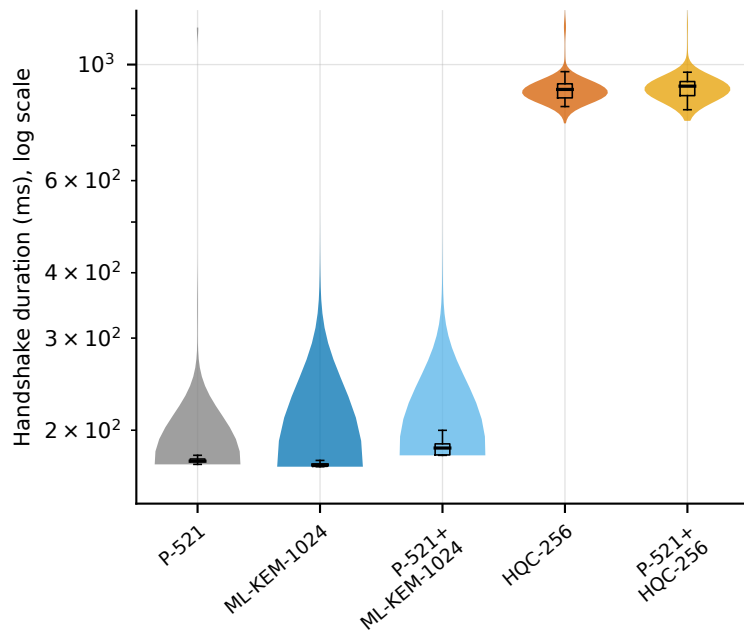
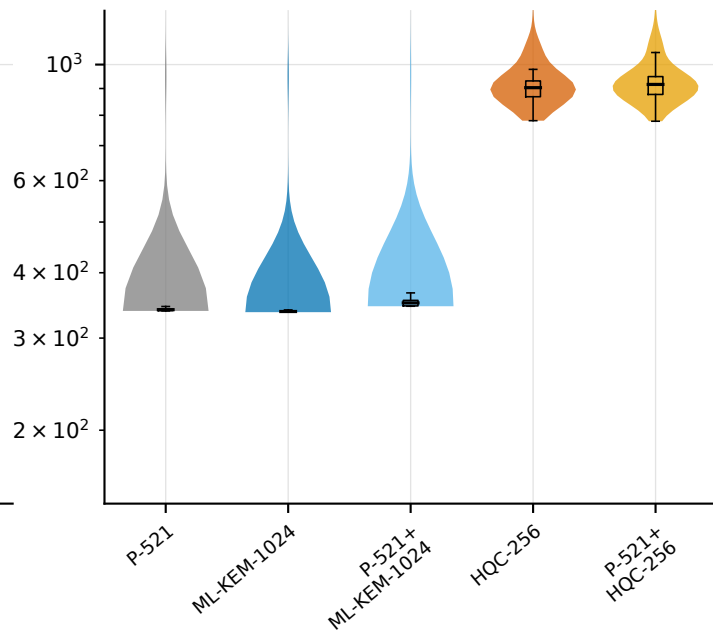


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	192.2	65.7	1.18k	382.3	83.4	3.86k
ML-KEM-1024	205.5	111.3	2.20k	380.0	74.5	2.67k
P-521+ML-KEM-1024	213.3	95.4	1.71k	415.8	88.4	3.06k
HQC-256	908.1	12.6	1.86k	952.9	22.8	3.62k
P-521+HQC-256	923.4	15.4	2.01k	961.9	20.3	2.64k